

Impact of pre-Columbian “geoglyph” builders on Amazonian forests

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Over 450 pre-Columbian (pre-AD 1492) geometric ditched enclosures (“geoglyphs”) occupy ~13,000 km² of Acre state, Brazil, representing a key discovery of Amazonian archaeology. These huge earthworks were concealed for centuries under *terra firme* (upland interfluvial) rainforest, directly challenging the “pristine” status of this ecosystem and its perceived vulnerability to human impacts. We reconstruct the environmental context of geoglyph construction and the nature, extent, and legacy of associated human impacts. We show that bamboo forest dominated the region for ≥6,000 y and that only small, temporary clearings were made to build the geoglyphs; however, construction occurred within anthropogenic forest that had been actively managed for millennia. In the absence of widespread deforestation, exploitation of forest products shaped a largely forested landscape that survived intact until the late 20th century.

Amazonian archaeology | Amazonian rainforest | paleoecology | pre-Columbian land use

The notion of Amazonia as a pristine wilderness has now been overturned by increasing evidence for large, diverse, and socially complex pre-Columbian societies in many regions of the basin. The discovery of numerous, vast *terra preta* (anthropogenic dark earth) sites bordering the floodplains of major rivers, and extensive earthwork complexes in the seasonally flooded savannas of the Llanos de Mojos (northeast Bolivia), Marajó Island (northeast Brazil), and coastal French Guiana, are seen to represent examples of major human impacts carried out in these environments (1–10).

However, major disagreement still resides in whether interfluvial forests, which represent over 90% of Amazonian ecosystems, were settings of limited, temporary human impacts (11–13), or were instead extensively transformed by humans over the course of millennia (14–16). A paucity of paleoecological studies conducted in interfluvial areas has been responsible for the polarization of this debate, which encompasses different hypothetical estimates of precontact population size and carrying capacity in the interfluvies (17), and the relative importance of different land use strategies in the past. The extent of ancient forest burning is particularly contested, because some have proposed that pre-Columbian deforestation was on a large enough scale to have influenced the carbon cycle and global climate (18, 19), whereas others argue that large-scale slash-and-burn agriculture is a largely postcontact phenomenon (20). Modern indigenous groups often subject slash-and-burn plots for crop cultivation to long fallow periods, during which useful plants, including many tree species, continue to be encouraged and managed in different stages of succession within a mosaic-type landscape (21, 22). Also known as “agroforestry,” this type of land use is thought to have been common in pre-Columbian times, but its detection in the paleoecological record is often problematic (15) and studies based on modern distributions of useful species lack demonstrable time

depth of forest modifications (23). Terrestrial paleoecology programs are essential for a better understanding of these issues, which have strong implications for the resilience of Amazonian forests to human impact and, subsequently, their future conservation (24–26).

With ditches up to 11 m wide, 4 m deep, and 100–300 m in diameter, and with some sites having up to six enclosures, the geoglyphs of western Amazonia rival the most impressive examples of pre-Columbian monumental architecture anywhere in the Americas (27). Excavations of the geoglyphs have shown that they were built and used sporadically as ceremonial and public gathering sites between 2000 and 650 calibrated years before present (BP), but that some may have been constructed as early as 3500–3000 BP (28–30). Evidence for their ceremonial function is based on an almost complete absence of cultural material found within the enclosed areas, which suggests they were kept ritually “clean,” alongside their highly formalized architectural forms (mainly circles and squares)—features that distinguish the geoglyphs from similar ditched enclosures in northeast Bolivia (5, 31). Surprisingly, little is known about who the geoglyph builders were and how and where they lived, as contemporary settlement sites have not yet been found in the region. It is thought that the geoglyph builders were a complex network of local, relatively autonomous groups connected by a shared and

Significance

Amazonian rainforests once thought to be pristine wildernesses are increasingly known to have been inhabited by large populations before European contact. How and to what extent these societies impacted their landscape through deforestation and forest management is still controversial, particularly in the vast interfluvial uplands that have been little studied. In Brazil, the groundbreaking discovery of hundreds of geometric earthworks by modern deforestation would seem to imply that this region was also deforested to a large extent in the past, challenging the apparent vulnerability of Amazonian forests to human land use. We reconstructed environmental evidence from the geoglyph region and found that earthworks were built within man-made forests that had been previously managed for millennia. In contrast, long-term, regional-scale deforestation is strictly a modern phenomenon.

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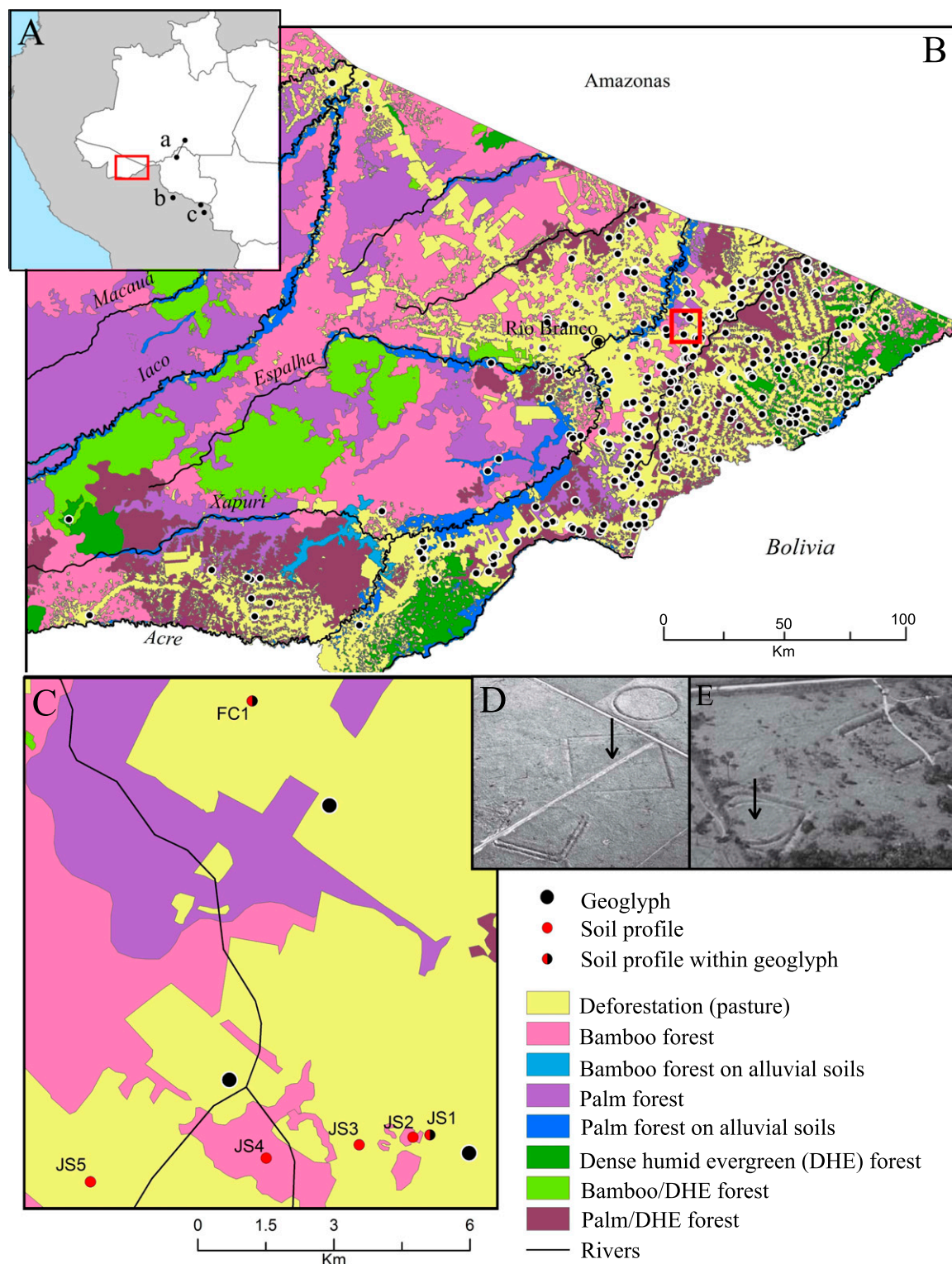


Fig. 1. Location maps. (A) Location of eastern Acre (*Inset*) and paleoecological studies that document a savanna–rainforest transition in the mid-Holocene: (a) soil profiles between Porto Velho and Humaitá (stable carbon isotopes), (b) Lagunas Oricore and Granja (pollen), and (c) Lagunas Bella Vista and Chaplin (pollen). (B) Geoglyph distributions in relation to modern vegetation and location of FC and JS sites (*Inset*). (C) Locations of soil profiles in this study. From the JS geoglyph (0 km), profile distances along the transect are as follows: JS2 = 0.5 km, JS3 = 1.5 km, JS4 = 3.5 km, and JS5 = 7.5 km. (D and E) Aerial photos of the FC (D) and JS (E) geoglyphs. Black arrows show the locations of profiles FC1 and JS1.

localized. This suggests that the geoglyphs were not designed for intervisibility, but were instead hidden from view: an unexpected conclusion.

Rather than being built within largely “untouched” bamboo forest, our phytolith data suggest that the geoglyphs were constructed within anthropogenic forests that had already been fundamentally altered

surface-soil phytolith assemblages from this forest patch plotted close to phytolith sample 20–25 cm in the JS2 profile (Fig. S2), immediately below the peak in palm phytoliths, implying that legacies of pre-Columbian agroforestry still exist today within Acre's remaining forests.

Implications

In contrast to studies that argue for either minimal (11, 12) or widespread (15, 16) pre-Columbian impact on the Amazonian interfluvies, we suggest that, in Acre, geoglyph construction was not associated with deforestation over large spatial and temporal scales but instead with a long tradition of agroforestry and resource management that altered the composition of native bamboo forest over millennia.

Our findings challenge the hypothesis that reforestation after the Columbian encounter led to a sequestration of CO₂ that triggered the Little Ice Age global cooling event (18, 19). This hypothesis was formerly criticized in light of findings that many earthworks in northeast Bolivia were constructed in nonforested landscapes (26), but our data indicate that, even in an archaeologically rich area that remained forested during the mid- to late Holocene, pre-Columbian deforestation was on a more localized scale than previously thought. Despite the number and density of geoglyphs, we did not find any pre-Columbian parallel for the length and extent of modern-day forest clearance in Acre.

Our data also raise a methodological concern crucial to the interpretation of terrestrial paleoecological data—namely, that low soil charcoal frequencies do not necessarily correlate with sparse pre-Columbian populations in Amazonia (11). There is little question that the geoglyphs are a product of sizeable, socially complex societies that once inhabited the region (27, 32), so the absence of evidence of large-scale deforestation in our soil profiles casts doubt over whether quantification of forest burning should play such a central role in delimiting areas of high vs. low populations, and minimal vs. widespread environmental impacts associated with them.

In contrast, our study has provided empirical, paleoecological evidence for the importance of forest management practices in the pre-Columbian interfluvies. The proliferation of palms and other useful species over apparently millennial timescales suggests a long history of forest manipulation before the JS and FC geoglyphs were even constructed, consistent with some arguments that long-term accumulations of small-scale disturbances can fundamentally alter species composition (15, 16).

We did not detect anthropogenic forest in all profile locations but recognize that formations not rich in palms are currently very difficult to detect in the phytolith record. This point is made clear by the species and phytolith data from the JS2 forest plot, which hint at the other species that were favored by pre-Columbian populations (e.g., Brazil nut) that do not produce diagnostic phytoliths.

We have shown that at least some of Acre's surviving forest owes its composition to sustainable pre-Columbian forest management practices that, combined with short-term, localized deforestation, maintained a largely forested landscape until the mid-20th century. The lack of a pre-Columbian analog for extensive modern deforestation means that we should not assume forest resilience to this type of land use, nor its recovery in the future.

Methods

Soil profiles were dug to 1.5-m depth and sampled in 5-cm increments for paleoecological analyses. Chronologies were based upon four accelerator mass spectrometry (AMS) dates per soil profile, the majority being determined on bulk macroscopic (>125 μ m) charcoal (19 samples), and the remainder on soil humin (6 samples) (Table S1). Due to the occurrence of age inversions, the integrity of the proxy data was assessed based on interprofile replicability of observed patterns (e.g., increases in palm phytoliths) and obtained dates consistent with geoglyph chronologies (SI Text, Age Inversions). Phytoliths were extracted every 5 cm in levels pertaining to geoglyph use and every 10 cm thereafter, following the wet oxidation method (55). Two hundred morphotypes were identified per soil sample, and taxa were identified using published atlases and the University of Exeter phytolith reference collection. Paleoecological phytolith assemblages were compared with assemblages from surface soils of modern forests in the region (56). Charcoal was extracted using a macroscopic sieving method (57) and divided into size classes to distinguish local (>250 μ m) from extralocal (125–250 μ m) burning signals. Stable carbon isotope analysis of SOM was conducted at JS1 (every 10 cm) and JS3 (every 10 cm, then every 20 cm below 0.4 m below surface) using standard procedures (58). Detailed information for the methodologies used in this study is provided in SI Methods.

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- Denevan WM (1992) The pristine myth: The landscape of the Americas in 1492. *Ann Assoc Am Geogr* 82(3):369–385.
- Neves EG, Petersen JB (2006) Political economy and pre-Columbian landscape transformations in Central Amazonia. *Time and Complexity in Historical Ecology* (Columbia Univ Press, New York), pp 279–309.
- Heckenberger M, Neves EG (2009) Amazonian archaeology. *Annu Rev Anthropol* 38(1):251–266.
- McMichael CH, et al. (2014) Predicting pre-Columbian anthropogenic soils in Amazonia. *Proc Biol Sci* 281(1777):20132475.
- Denevan WM (1966) *The Aboriginal Cultural Geography of the Llanos de Mojos of Bolivia*. Ibero-Americana 48 (University of California Press, Berkeley, CA).
- Erickson CL (2006) The domesticated landscapes of the Bolivian Amazon. *Time and Complexity in Historical Ecology*, eds Erickson CL, Balée W (Columbia Univ Press, New York), pp 235–278.
- Erickson CL (2000) The Lake Titicaca basin: A pre-Columbian built landscape. *Imperfect Balance: Landscape Transformations in the Pre-Columbian Americas*, ed Lentz D (Columbia Univ Press, New York), pp 311–356.
- Schaan D (2008) The non-agricultural chiefdoms of Marajó Island. *Handbook of South American Archaeology*, eds Silverman H, Isbell WH (Springer, New York), pp 339–357.
- Rostain S (2010) Pre-Columbian earthworks in coastal Amazonia. *Diversity (Basel)* 2(3): 331–352.
- Iriarte J, et al. (2010) Late Holocene neotropical agricultural landscapes: Phytolith and stable carbon isotope analysis of raised fields from French Guianan coastal savannahs. *J Archaeol Sci* 37(12):2984–2994.
- McMichael CH, et al. (2012) Sparse pre-Columbian human habitation in western Amazonia. *Science* 336(6087):1429–1431.
- Piperno DR, McMichael C, Bush MB (2015) Amazonia and the Anthropocene: What was the spatial extent and intensity of human landscape modification in the Amazon Basin at the end of prehistory? *Holocene* 25(10):1588–1597.
- Bush MB, et al. (2015) Anthropogenic influence on Amazonian forests in pre-history: An ecological perspective. *J Biogeogr* 42(12):2277–2288.
- Roosevelt AC (2014) The Amazon and the Anthropocene: 13,000 years of human influence in a tropical rainforest. *Anthropocene* 4:69–87.
- Clement CR, et al. (2015) The domestication of Amazonia before European conquest. *Proc Biol Sci* 282(1812):20150813.
- Stahl PW (2015) Interpreting interfluvial landscape transformations in the pre-Columbian Amazon. *Holocene* 25(10):1598–1603.
- Denevan WM (2014) Estimating Amazonian Indian numbers in 1492. *J Lat Am Geogr* 13(2):207–221.
- Neville RJ, Bird DK (2008) Effects of syn-pandemic fire reduction and reforestation in the tropical Americas on atmospheric CO₂ during European conquest. *Palaeogeogr Palaeoclimatol Palaeoecol* 264(1–2):25–38.
- Dull RA, et al. (2010) The Columbian Encounter and the Little Ice Age: Abrupt land use change, fire, and greenhouse forcing. *Ann Assoc Am Geogr* 100(4):37–41.
- Arroyo-Kalin M (2012) Slash-burn-and-churn: Landscape history and crop cultivation in pre-Columbian Amazonia. *Quat Int* 249:4–18.
- Posey A, Balée W (1989) *Resource Management in Amazonia: Indigenous Folk Strategies* (New York Botanical Garden, New York).
- Clement CR (2006) Demand for classes of traditional agroecological knowledge in modern Amazonia. *Human Impacts on Amazonia: The Role of Traditional Ecological Knowledge in Conservation and Development*, eds Posey DA, Balick MJ (Columbia Univ Press, New York), pp 33–50.
- Clement CR, et al. (2015) Response to comment by McMichael, Piperno and Bush. *Proc Biol Sci* 282(1821):20152459.

24. Heckenberger MJ, Russell JC, Toney JR, Schmidt MJ (2007) The legacy of cultural landscapes in the Brazilian Amazon: Implications for biodiversity. *Philos Trans R Soc Lond B Biol Sci* 362(1478):197–208.
25. Bush MB, Silman MR (2007) Amazonian exploitation revisited: Ecological asymmetry and the policy pendulum. *Front Ecol Environ* 5(9):457–465.
26. Carson JF, et al. (2014) Environmental impact of geometric earthwork construction in pre-Columbian Amazonia. *Proc Natl Acad Sci USA* 111(29):10497–10502.
27. Schaan D (2012) *Sacred Geographies of Ancient Amazonia: Historical Ecology of Social Complexity* (Left Coast Press, San Francisco).
28. Saunaluoma S, Schaan D (2012) Monumentality in western Amazonian formative societies: Geometric ditched enclosures in the Brazilian state of Acre. *Antiqua* 2:e1.
29. Schaan D, et al. (2012) New radiometric dates for precolumbian (2000–700 B.P.) earthworks in western Amazonia, Brazil. *J Field Archaeol* 37(2):132–142.
30. Pärssinen M, Schaan D, Ranzi A (2009) Pre-Columbian geometric earthworks in the upper Purus: A complex society in western Amazonia. *Antiquity* 83(322):1084–1095.
31. Prúmers H (2014) Sitios prehispánicos con zanjas en Bella Vista, Provincia Iténez, Bolivia. *Antes de Orellana. Actas Del 3er Encuentro Internacional de Arqueología Amazónica*, ed Rostain S (IFEA; FLASCO; MCCTH; SENESCYT, Quito, Ecuador), pp 73–91.
32. Saunaluoma S, Virtanen PK (2015) Variable models for organization of earthworking communities in Upper Purus, southwestern Amazonia: Archaeological and ethnographic perspectives. *Tipiti J Soc Anthropol Lowl South Am* 13(1):23–43.
33. Eriksen L (2011) Nature and culture in prehistoric Amazonia. PhD dissertation (Lund University, Lund, Sweden).
34. Saunaluoma S (2016) Cerâmicas do Acre. *Cerâmicas Arqueológicas Da Amazônia: Rumo a Uma Nova Síntese*, eds Barreto C, Lima HP, Betancourt CJ (IPHAN: Ministério de Cultura, Belém, PA, Brazil), pp 414–419.
35. de Carvalho AL, et al. (2013) Bamboo-dominated forests of the southwest Amazon: Detection, spatial extent, life cycle length and flowering waves. *PLoS One* 8(1):e54852.
36. McMichael CH, Palace MW, Golightly M (2014) Bamboo-dominated forests and pre-Columbian earthwork formations in south-western Amazonia. *J Biogeogr* 41(9):1733–1745.
37. McMichael CH, et al. (2013) Historical fire and bamboo dynamics in western Amazonia. *J Biogeogr* 40(3):299–309.
38. Olivier J, et al. (2009) First macrofossil evidence of a pre-Holocene thorny bamboo cf. *Guadua* (Poaceae: Bambusoideae: Bambuseae: Guaduiniae) in south-western Amazonia (Madre de Dios–Peru). *Rev Palaeobot Palynol* 153:1–7.
39. Mayle FE, Power MJ (2008) Impact of a drier Early-Mid-Holocene climate upon Amazonian rain forests. *Philos Trans R Soc Lond B Biol Sci* 363(1498):1829–1838.
40. Carson JF, et al. (2015) Pre-Columbian land use in the ring-ditch region of the Bolivian Amazon. *Holocene* 25(8):1285–1300.
41. Duarte AF (2005) Variabilidade e tendência das chuvas em Rio Branco, Acre, Brasil. *Rev Bras Meteorol* 20:37–42.
42. Aragão LEOC, et al. (2008) Interactions between rainfall, deforestation and fires during recent years in the Brazilian Amazonia. *Philos Trans R Soc Lond B Biol Sci* 363(1498):1779–1785.
43. Daly DC, Silveira M (2008) *First Catalogue of the Flora of Acre* (Universidade Federal do Acre, Rio Branco, Brazil).
44. Quesada CA, et al. (2011) Soils of Amazonia with particular reference to the RAINFOR sites. *Biogeosciences* 8(6):1415–1440.
45. Baker PA, et al. (2001) The history of South American tropical precipitation for the past 25,000 years. *Science* 291(5504):640–3.
46. Mayle FE, Burbridge R, Killeen TJ (2000) Millennial-scale dynamics of southern Amazonian rain forests. *Science* 290(5500):2291–2294.
47. Pessenda LCR, et al. (2001) Origin and dynamics of soil organic matter and vegetation changes during the Holocene in a forest-savanna transition zone, Brazilian Amazon region. *Holocene* 11(2):250–254.
48. de Freitas HA, et al. (2001) Late Quaternary vegetation dynamics in the southern Amazon basin inferred from carbon isotopes in soil organic matter. *Quat Res* 55(1):39–46.
49. Watling J, Saunaluoma S, Pärssinen M, Schaan D (2015) Subsistence practices among earthwork builders: Phytolith evidence from archaeological sites in the southwest Amazonian interfluvies. *J Archaeol Sci Reports* 4:541–551.
50. Dickau R, et al. (2013) Differentiation of neotropical ecosystems by modern soil phytolith assemblages and its implications for palaeoenvironmental and archaeological reconstructions. *Rev Palaeobot Palynol* 193:15–37.
51. Clement CR (1999) 1492 and the loss of Amazonian crop genetic resources. I: The relation between domestication and human population decline. *Econ Bot* 53(2):188–202.
52. Kahn F (1988) Ecology of economically important palms in Amazonia. *Adv Econ Bot* 6:42–49.
53. Kahn F, Granville JJ (1992) Palms in forest ecosystems of Amazonia. *Ecological Studies* (Springer, Berlin), Vol 95, pp 41–89.
54. Salm R, Jalles-filho E, Schuck-paim C (2005) A model for the importance of large arborescent palms in the dynamics of seasonally-dry Amazonian forests. *Biota Neotropica* 5(1):1–6.
55. Piperno DR (2006) *Phytoliths: A Comprehensive Guide for Archaeologists and Paleoecologists* (Altamira, Oxford, UK).
56. Watling J, et al. (2016) Differentiation of neotropical ecosystems by modern soil phytolith assemblages and its implications for palaeoenvironmental and archaeological reconstructions. II: Southwestern Amazonian forests. *Rev Palaeobot Palynol* 226:30–43.
57. Whitlock C, Larsen C (2001) Charcoal as a fire proxy. *Tracking Environmental Change Using Lake Sediments, Volume 3: Terrestrial, Algal and Siliceous Indicators*, eds Smol JP, Birks HJB, Last WM (Kluwer Academic, Dordrecht, The Netherlands), pp 75–97.
58. Metcalfe SE, et al. (2014) Hydrology and climatology at Laguna La Gaiba, lowland Bolivia: Complex responses to climatic forcings over the last 25,000 years. *J Quat Sci* 29(3):289–300.
59. Morcote-Rios G, Bernal R, Raz L (2016) Phytoliths as a tool for archaeobotanical, palaeobotanical and palaeoecological studies in Amazonian palms. *Bot J Linn Soc* 182(2):348–360.
60. Clark JS (1988) Particle motion and the theory of charcoal analysis: Source area, transport, deposition and sampling. *Quat Res* 30:67–80.
61. Gardner JJ, Whitlock C (2001) Charcoal accumulation following a recent fire in the Cascade Range, northwestern USA, and its relevance for fire-history studies. *Holocene* 11(5):541–549.
62. Kelly E, et al. (1998) Stable isotope composition of soil organic matter and phytoliths as paleoenvironmental indicators. *Geoderma* 82(1-3):59–81.
63. Pessenda LCR, et al. (1998) The carbon isotope record in soils along a forest-cerrado ecosystem transect: Implications for vegetation changes in the Rondonia state, southwestern Brazilian Amazon region. *Holocene* 8(5):599–603.
64. Pessenda LCR, Camargo PB (1991) Datação radiocarbônica de amostras de interesse arqueológico e geológico por espectrometria de cintilação líquida de baixa radiação de fundo. *Quim Nova* 2:98–103.
65. Pessenda LCR, Gouveia SEM, Aravena R (2001) Radiocarbon dating of total soil organic matter and humin fraction and its comparison with ^{14}C ages of fossil charcoal. *Radiocarbon* 43(2B):595–601.
66. Soubiès F (1979) Existence d'une phase sèche en Amazonie Brésilienne datée par la présence de charbons dans les sols (6.000-8.000 ans B.P.). *Cah ORSTOM Ser Geol* 9(1):133–146.
67. Piperno DR, Becker P (1996) Vegetational history of a site in the central Amazon basin derived from phytolith and charcoal records from natural soils. *Quat Res* 45(2):202–209.
68. Zangerlé A, et al. (2016) The Surales, self-organized earth-mound landscapes made by earthworms in a seasonal tropical wetland. *PLoS One* 11(5):e0154269.
69. Twiss PC, Suess E, Smith RM (1969) Morphological classifications of grass phytoliths. *Soil Sci Soc Am J* 33(1):109–115.
70. Piperno DR, Pearsall DM (1998) The silica bodies of tropical American grasses: Morphology, taxonomy, and implications for grass systematics and fossil phytolith identification. *Smithson Contrib Bot* 85:1–40.
71. Geis JW (1973) Biogenic silica in selected species of deciduous angiosperms. *Soil Sci* 116:113–130.
72. Wilding LP, Drees LR (1974) Contributions of forest opal and associated crystalline phases to fine silt and clay fractions of soils. *Clays Clay Miner* 22:295–306.
73. Bremond L, Alexandre A, Hely C, Guiot J (2005) A phytolith index as a proxy of tree cover density in tropical areas: Calibration with leaf area index along a forest-savanna transect in southeastern Cameroon. *Global Planet Change* 45(4):277–293.
74. Runge F (1999) The opal phytolith inventory of soils in central Africa—quantities, shapes, classification, and spectra. *Rev Palaeobot Palynol* 107(1-2):23–53.
75. Strömberg CE (2004) Using phytolith assemblages to reconstruct the origin and spread of grass-dominated habitats in the great plains of North America during the late Eocene to early Miocene. *Palaeogeogr Palaeoclimatol Palaeoecol* 207(3-4):239–275.
76. Sanaiotti ATM, et al. (2002) Past vegetation changes in Amazon savannas determined using carbon isotopes of soil organic matter. *Biotropica* 34(1):2–16.
77. Reimer PJ, et al. (2013) IntCal13 and Marine13 radiocarbon age calibration curves, 0–50,000 years cal BP. *Radiocarbon* 55(4):1869–1887.
78. McCormac FG, et al. (2004) SHCal04 Southern hemisphere calibration, 0–11.0 cal kyr BP. *Radiocarbon* 46(3):1087–1092.